

Mathematic Foundation and Basic Concept

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Recap: Prerequisites

- Basic knowledge of probabilities
 - Bayes rule, chain rule, probability distribution ...
- Basic knowledge of information theory
 - Self-information, Shannon entropy, differential entropy ...
 - Kullback-Leibler (KL) divergence
- Basic knowledge of machine learning/deep learning
 - “Machine Learning”, “Pattern Recognition and Machine Learning”
 - “Computer Vision”, “Natural Language Processing” ...
- Basic programming language
 - Python

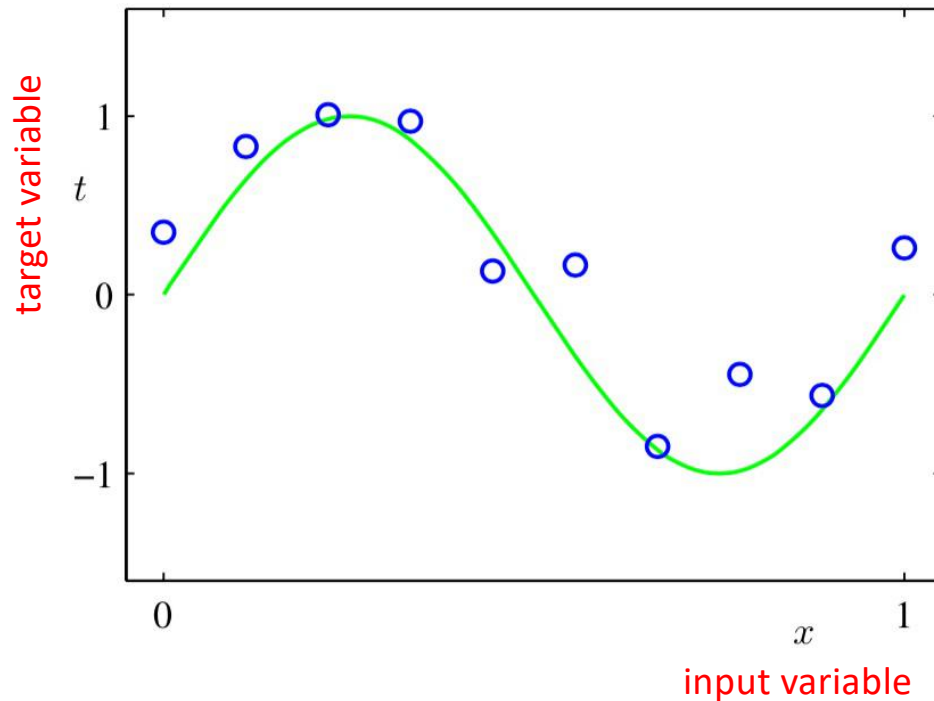
Mathematic Foundation and Basic Concept

- Example: Regression - Polynomial Curve Fitting
- Probability Theory
- Decision Theory
- Information Theory

- **Example: Regression - Polynomial Curve Fitting**
- Probability Theory
- Decision Theory
- Information Theory

Example: Polynomial Curve Fitting

- Problem Definition

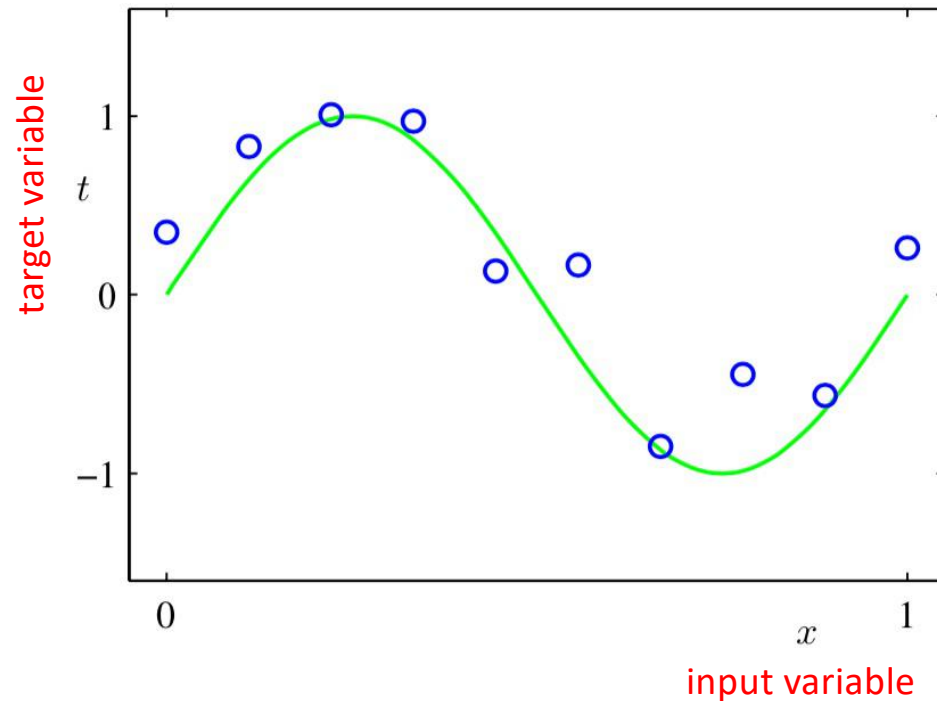


- A training dataset of $N = 10$ points
- Predict the target value t given the input x

$\mathbf{x} \equiv (x_1, \dots, x_N)^T$ A row vector with N elements
 $\mathbf{t} \equiv (t_1, \dots, t_N)^T$

Example: Polynomial Curve Fitting

- Simple Model: Polynomial Function



M order polynomial:

$$y(x, \mathbf{w}) = w_0 + w_1x + w_2x^2 + \dots + w_Mx^M = \sum_{j=0}^M w_jx^j$$

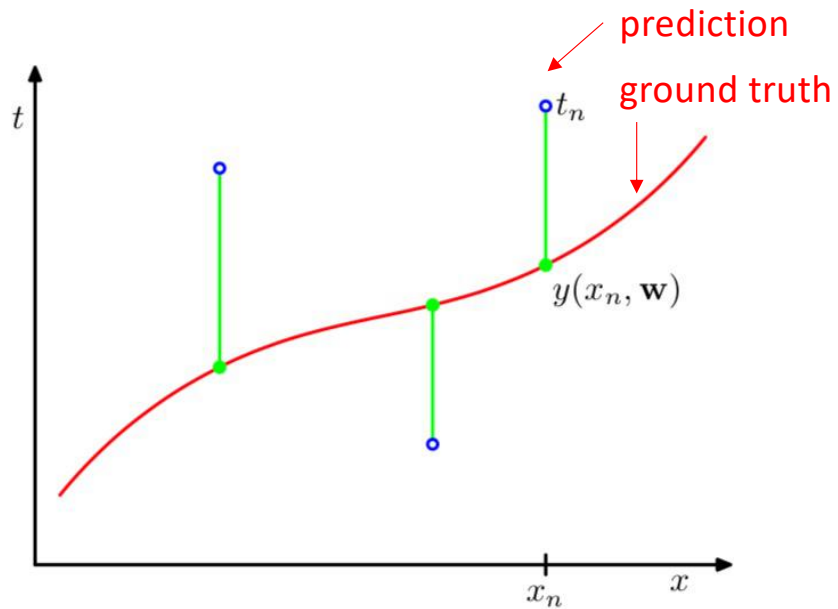
$\mathbf{w} = (w_1, \dots, w_M)^T$ **Weights:** a row vector with M elements

The appropriate value $\hat{t}$:

$$\hat{t} = \mathbf{x}^T \mathbf{w}$$

Example: Polynomial Curve Fitting

- Error Function



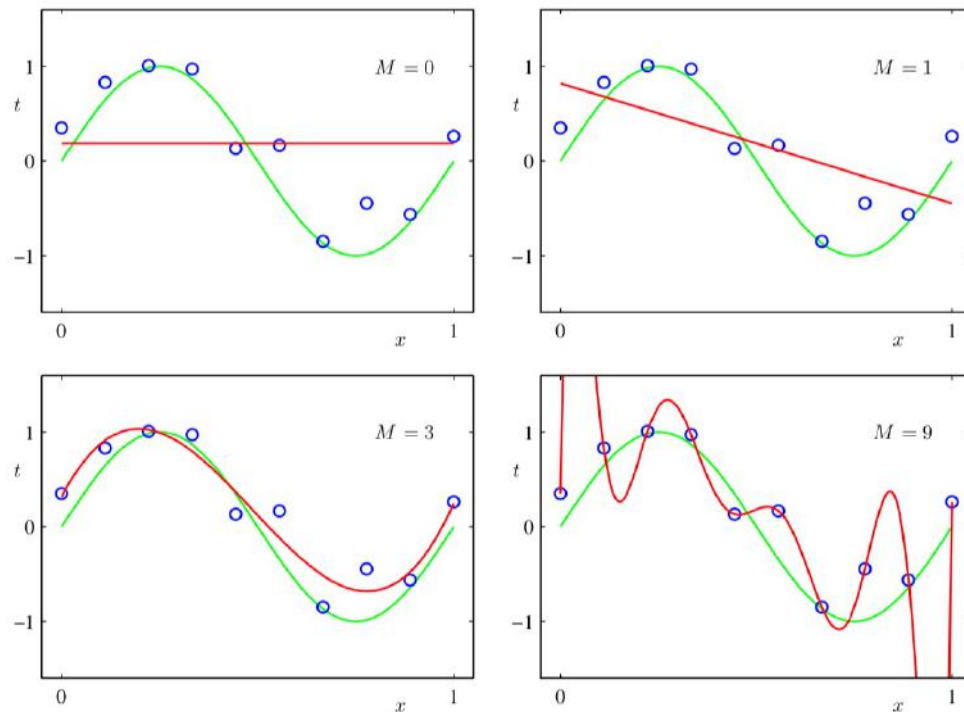
Minimise an error function

$$E(\mathbf{w}) = \frac{1}{2} \sum_{n=1}^N \{y(x_n, \mathbf{w}) - t_n\}^2$$

Optimise the weights

Example: Polynomial Curve Fitting

- Model Capacity and Overfitting



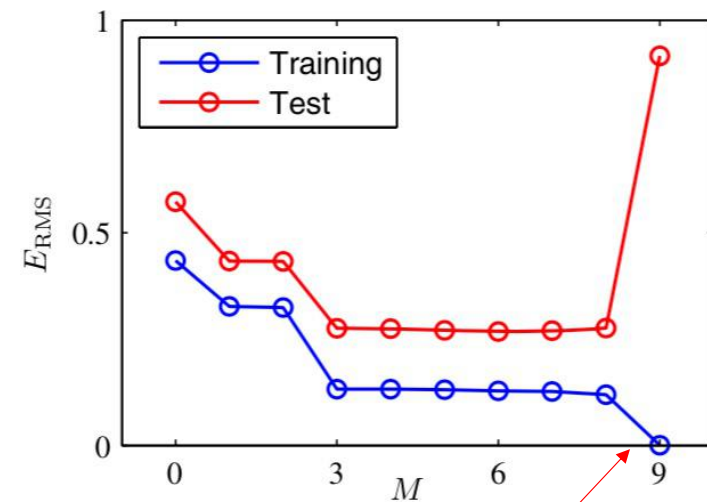
M order polynomial

Root mean square error:

$$E_{\text{RMS}} = \sqrt{2E(\mathbf{w}^*)/N}$$

$$E(\mathbf{w}^*) = 0$$

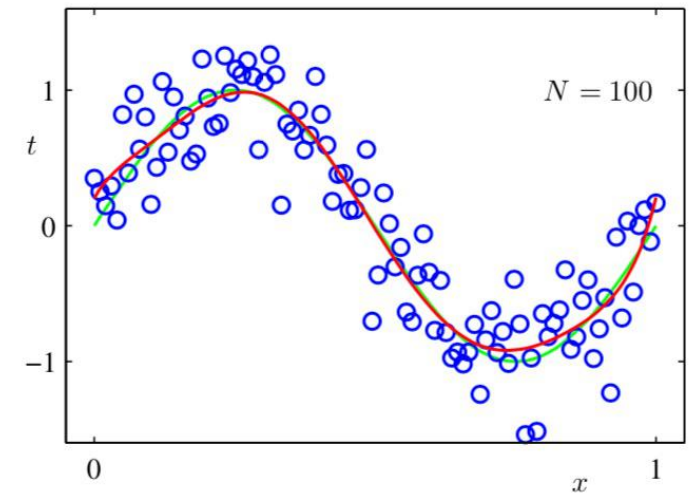
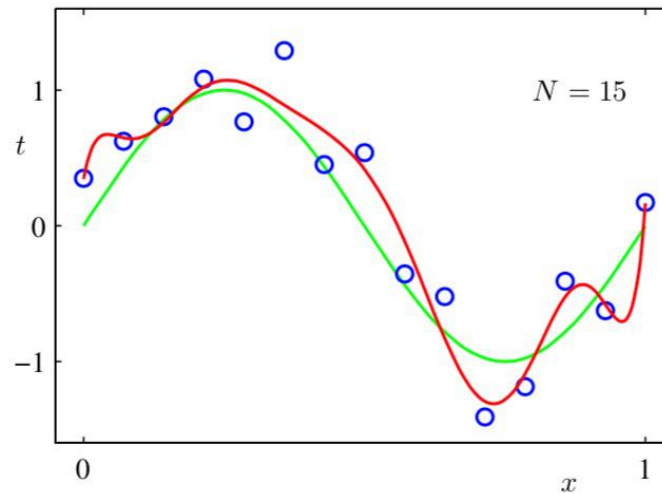
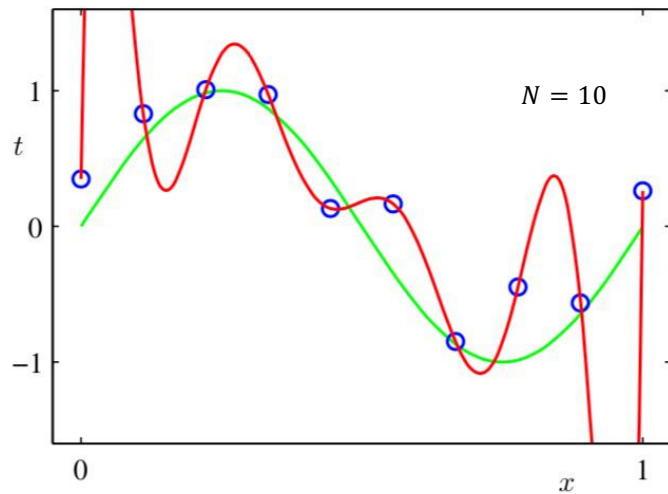
Averaged/mean error
Optimised weights



Large M will lead to overfitting

Example: Polynomial Curve Fitting

- Solution1 : More Training Data Point



$M = 9$ with $N = 10, 15, 100$
More data points = Better generalization

Example: Polynomial Curve Fitting

- Solution2 : Weight Regularisation

	$M = 0$	$M = 1$	$M = 6$	$M = 9$
w_0^*	0.19	0.82	0.31	0.35
w_1^*		-1.27	7.99	232.37
w_2^*			-25.43	-5321.83
w_3^*			17.37	48568.31
w_4^*				-231639.30
w_5^*				640042.26
w_6^*				-1061800.52
w_7^*				1042400.18
w_8^*				-557682.99
w_9^*				125201.43

↑
Optimised weights

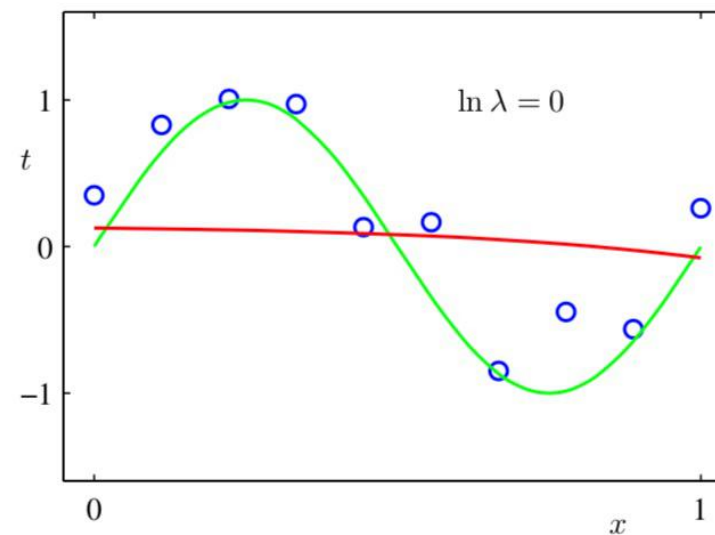
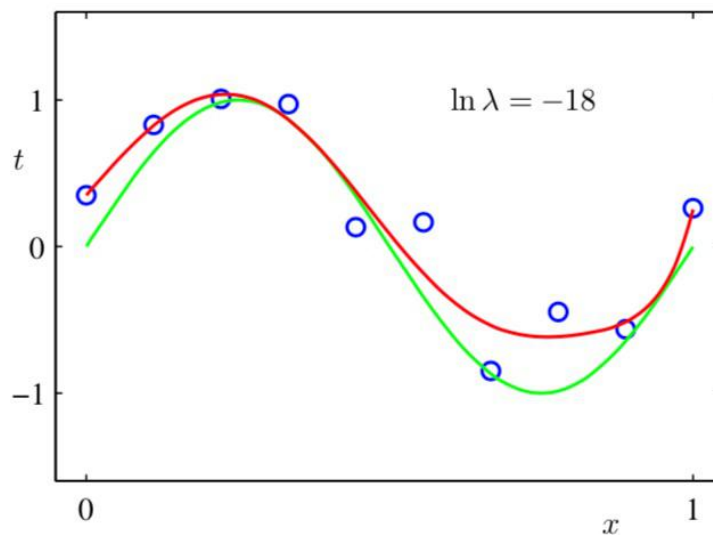
Very large

Example: Polynomial Curve Fitting

- Solution2 : Weight Regularisation

$$\tilde{E}(\mathbf{w}) = \frac{1}{2} \sum_{n=1}^N \{y(x_n, \mathbf{w}) - t_n\}^2 + \frac{\lambda}{2} \|\mathbf{w}\|^2$$

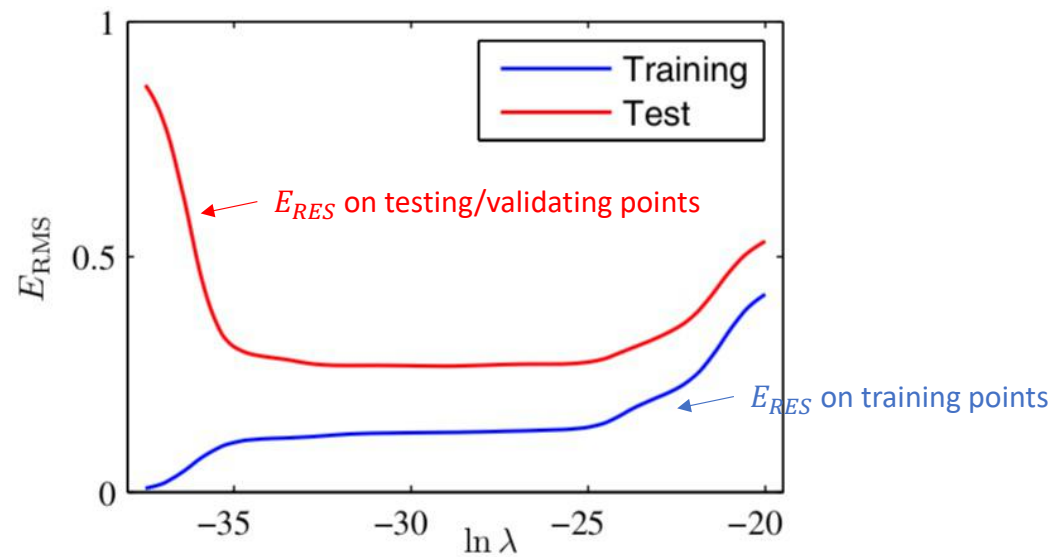
$$\|\mathbf{w}\|^2 \equiv \mathbf{w}^T \mathbf{w} = w_0^2 + w_1^2 + \dots + w_M^2$$



$M = 9$, with different λ
Larger λ = Stronger regularization

Example: Polynomial Curve Fitting

- Solution2 : Weight Regularisation



$M = 9$, with different λ
Larger λ = Stronger regularization

- Example: Regression - Polynomial Curve Fitting
- **Probability Theory**
- Decision Theory
- Information Theory

Probability Theory

- The probability of an event is the fraction of times that event occurs out of the total number of trials
- The probability must lie in the interval $[0, 1]$
- A box contains red and blue balls, we randomly pick 100 balls from a box, and 90 balls are red

$$p(ball = red) = \frac{90}{100} = 0.9$$

Probability Theory

y_j			n_{ij}	
			x_i	

Joint probability: $X = x_i$ and $Y = y_i$ happen together

$$p(X = x_i, Y = y_j) = \frac{n_{ij}}{N}$$

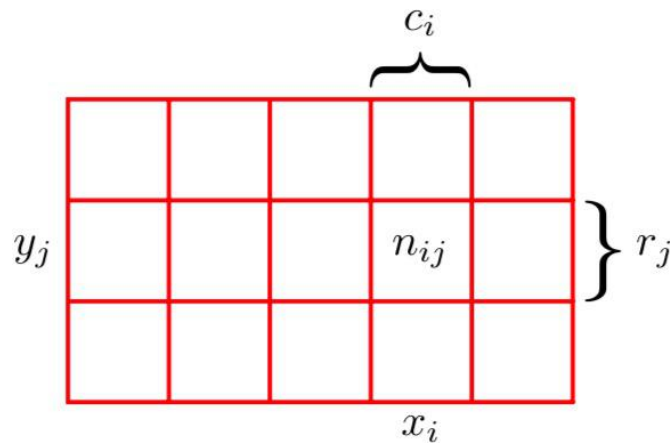
Given ..

$$p(X = x_i) = \frac{c_i}{N}$$

So.. We have the **sum rule of probability**

$$p(X = x_i) = \sum_{j=1}^L p(X = x_i, Y = y_j)$$

Probability Theory



Variable Value
↓ ↓

Conditional probability: $Y = y_i$ given $X = x_i$

$$p(Y = y_j | X = x_i) = \frac{n_{ij}}{c_i}$$

As we know:

$$p(X = x_i, Y = y_j) = \frac{n_{ij}}{N} \qquad p(X = x_i) = \frac{c_i}{N}$$

We can have the **product rule of probability**

$$\begin{aligned} p(X = x_i, Y = y_j) &= \frac{n_{ij}}{N} = \frac{n_{ij}}{c_i} \cdot \frac{c_i}{N} \\ &= p(Y = y_j | X = x_i) p(X = x_i) \end{aligned}$$

Probability Theory

- The Rules of Probability

sum rule $p(X) = \sum_Y p(X, Y)$

product rule $p(X, Y) = p(Y|X)p(X)$

Probability Theory

- Bayes' theorem

Given the **product rule**:

$$\begin{aligned} p(X, Y) &= p(Y|X)p(X) \\ p(Y, X) &= p(X|Y)p(Y) \end{aligned} \quad \begin{array}{c} \nearrow \\ \nwarrow \end{array} \text{symmetry property } p(X, Y) = p(Y, X)$$

We can have the **Bayes' theorem**:

$$p(Y|X) = \frac{p(X, Y)}{p(X)} = \frac{p(X|Y)p(Y)}{p(X)}$$

Using the **sum rule** ($p(X) = \sum_Y p(X, Y)$), we can have:

$$p(X) = \sum_Y p(X|Y)p(Y)$$

Probability Theory

- Bayes' theorem

The likelihood : It expresses how probable the observed data set is for different settings of the parameter y

The prior probability: it is the probability available before we observe the X

先验

似然

$$p(Y|X) = \frac{p(X|Y)p(Y)}{p(X)}$$

归一化常量

The normalisation constant required to ensure that the sum of the conditional probability over all values of Y equals one.

$$p(X) = \int_Y p(X|Y)p(Y)dY$$

The posterior probability: it is the probability obtained after we have observed X

后验

$\text{posterior} \propto \text{likelihood} \times \text{prior}$

Probability Theory

- The Rules of Probability

sum rule $p(X) = \sum_Y p(X, Y)$

product rule $p(X, Y) = p(Y|X)p(X)$

Bayes' theorem $p(Y|X) = \frac{p(X|Y)p(Y)}{p(X)}$

where $p(X) = \sum_Y p(X|Y)p(Y)$

Probability Theory

- Independent Events

If X and Y are **independent**

Bayes' theorem $p(Y|X) = \frac{p(X|Y)p(Y)}{p(X)}$

product rule $p(X, Y) = p(Y|X)p(X)$

$$p(Y|X) = p(Y)$$

$$p(X, Y) = p(X)p(Y)$$

Probability Theory

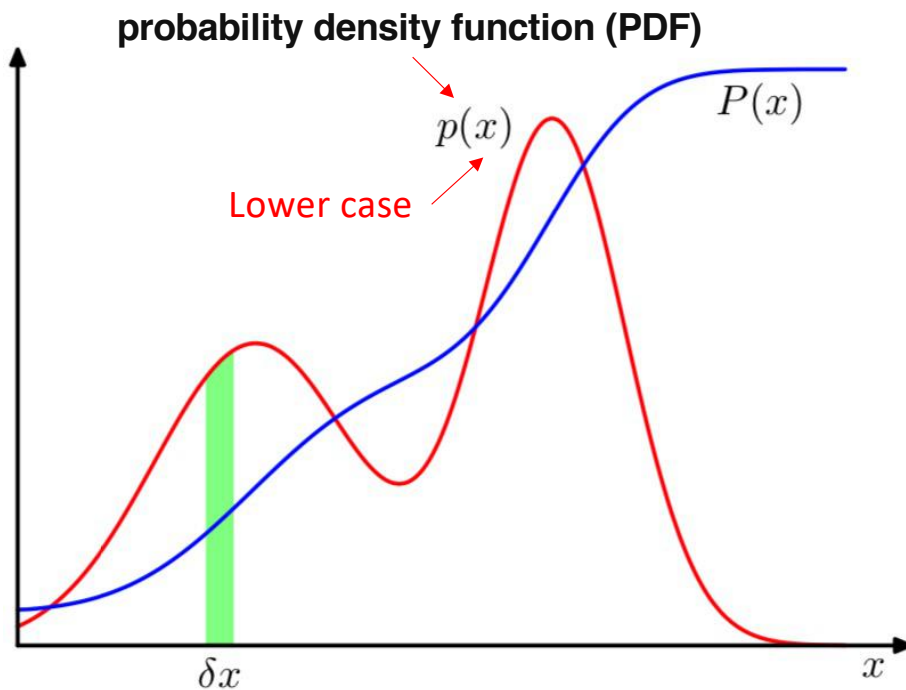
- Probability Densities

The probability that x will lie in an interval (a, b) is given by:

$$p(x \in (a, b)) = \int_a^b p(x) dx$$

Also ..

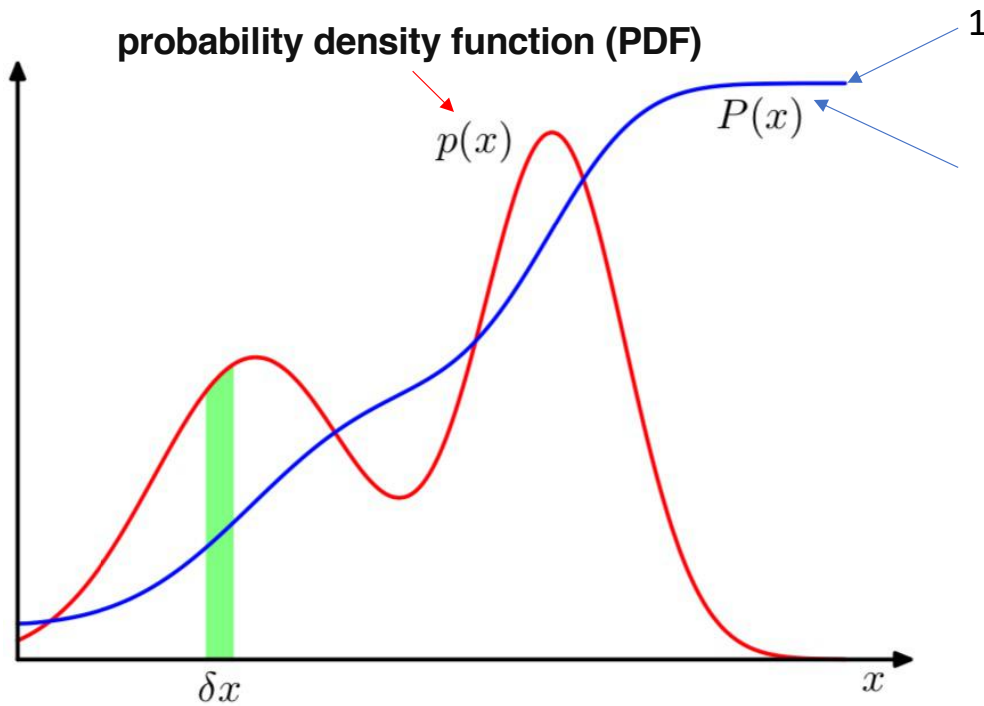
$$p(x) \geq 0$$
$$\int_{-\infty}^{\infty} p(x) dx = 1$$



If the probability of a real-valued variable x falling in the interval $(x, x + \delta x)$ is given by $p(x)\delta x$ for $\delta x \rightarrow 0$, then $p(x)$ is called **the probability density** over x .

Probability Theory

- Probability Densities



cumulative distribution function (CDF)

$$P(z) = \int_{-\infty}^z p(x) dx$$

which satisfies $P'(x) = p(x)$

Probability Theory

- **Expectations and Covariances**

The average value of some function $f(x)$ under a probability distribution $p(x)$ is called the **expectation** of $f(x)$ and will be denoted by $\mathbb{E}[f]$:

For a continuous distribution:

$$\mathbb{E}[f] = \int p(x) f(x) \, dx$$

For a discrete distribution:

$$\mathbb{E}[f] = \sum_x p(x) f(x)$$

$$\mathbb{E}[f] \simeq \frac{1}{N} \sum_{n=1}^N f(x_n)$$

It becomes exact in the limit $N \rightarrow \infty$

Probability Theory

- **Expectations and Covariances**

In the case of **continuous variables**, expectations are expressed in terms of an integration with respect to the corresponding probability density

$$\mathbb{E}[f] = \int p(x) f(x) \, dx$$

The conditional expectation with respect to a conditional distribution

$$\mathbb{E}_x[f|y] = \sum_x p(x|y) f(x)$$

Probability Theory

- Expectations and **Covariances**

The **variance** of $f(x)$:

$$\begin{aligned}\text{var}[f] &= \mathbb{E} [(f(x) - \mathbb{E}[f(x)])^2] \\ &= \mathbb{E}[f(x)^2] - \mathbb{E}[f(x)]^2\end{aligned}$$

The **variance** of the variable x :

$$\text{var}[x] = \mathbb{E}[x^2] - \mathbb{E}[x]^2$$

For two random variables x and y , the **covariance** is:

$$\begin{aligned}\text{cov}[x, y] &= \mathbb{E}_{x,y} [\{x - \mathbb{E}[x]\} \{y - \mathbb{E}[y]\}] \\ &= \mathbb{E}_{x,y}[xy] - \mathbb{E}[x]\mathbb{E}[y]\end{aligned}$$

For two vectors of random variables $\mathbf{x}$ and $\mathbf{y}$, the **covariance matrix**:

$$\begin{aligned}\text{cov}[\mathbf{x}, \mathbf{y}] &= \mathbb{E}_{\mathbf{x}, \mathbf{y}} [\{\mathbf{x} - \mathbb{E}[\mathbf{x}]\} \{\mathbf{y}^T - \mathbb{E}[\mathbf{y}^T]\}] \\ &= \mathbb{E}_{\mathbf{x}, \mathbf{y}} [\mathbf{x} \mathbf{y}^T] - \mathbb{E}[\mathbf{x}] \mathbb{E}[\mathbf{y}^T].\end{aligned}$$

The covariance of the components of a vector $\mathbf{x}$ with each other:

$$\text{cov}[\mathbf{x}] \equiv \text{cov}[\mathbf{x}, \mathbf{x}]$$

Probability Theory

- Bayesian Probabilities
 - **Frequentist:** Probability in terms of frequencies of random, repeatable events.
 - Do not work if the event is not repeatable, e.g., the probability of the Arctic ice cap disappear.
 - **Bayesian:** Degree of belief.

Probability Theory

- Frequentist Probability

- Maximum Likelihood Estimation, MLE

A widely used frequentist estimator is maximum likelihood, in which θ is set to the value that maximises the likelihood function $p(X|\theta)$.

- Given $x \sim \text{iid} \sim p(x|\theta)$, $p(X|\theta) = \prod_{i=1}^N p(x_i|\theta)$
 - $\theta_{\text{MLE}} = \arg \max_{\theta} \log p(X|\theta) = \arg \max_{\theta} \sum_{i=1}^N \log p(x_i|\theta)$

Probability Theory

- Bayesian Probabilities

- θ : random variables, $\theta \sim p(\theta) \leftarrow$ prior
- MAP: Maximum A Posterior

$$\underset{\text{posterior}}{p(\theta|X)} = \frac{\overset{\text{likelihood}}{p(X|\theta)}\overset{\text{prior}}{p(\theta)}}{\underset{\text{normalisation constant}}{p(X)}} \propto p(X|\theta)p(\theta)$$

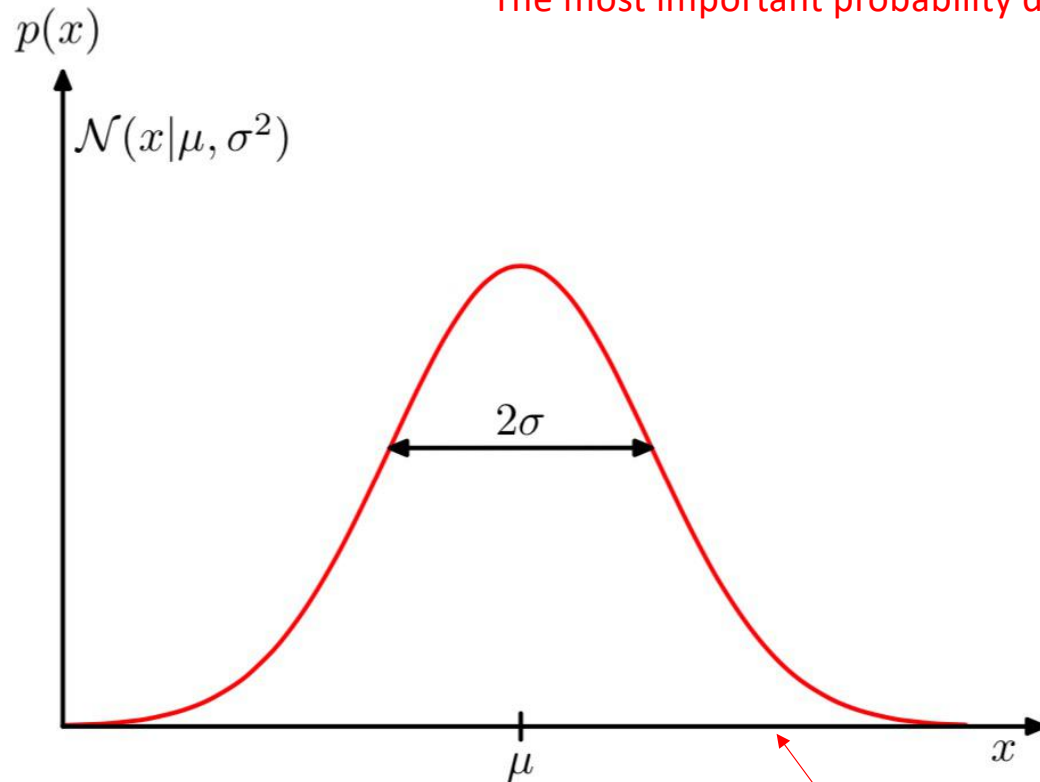
- $\theta_{\text{MAP}} = \arg \max_{\theta} p(\theta|X) = \arg \max_{\theta} p(X|\theta)p(\theta)$

Bayes can evaluate the **uncertainty** in θ *after* we have observed X in the form of the posterior probability $p(\theta|X)$.

Probability Theory

- The Gaussian Distribution

The most important probability distribution for continuous variables



$$\mathcal{N}(x|\mu, \sigma^2) = \frac{1}{(2\pi\sigma^2)^{1/2}} \exp \left\{ -\frac{1}{2\sigma^2} (x - \mu)^2 \right\}$$

$$\mathcal{N}(x|\mu, \sigma^2) > 0 \quad \int_{-\infty}^{\infty} \mathcal{N}(x|\mu, \sigma^2) dx = 1$$

μ : mean

σ : standard deviation

σ^2 : variance

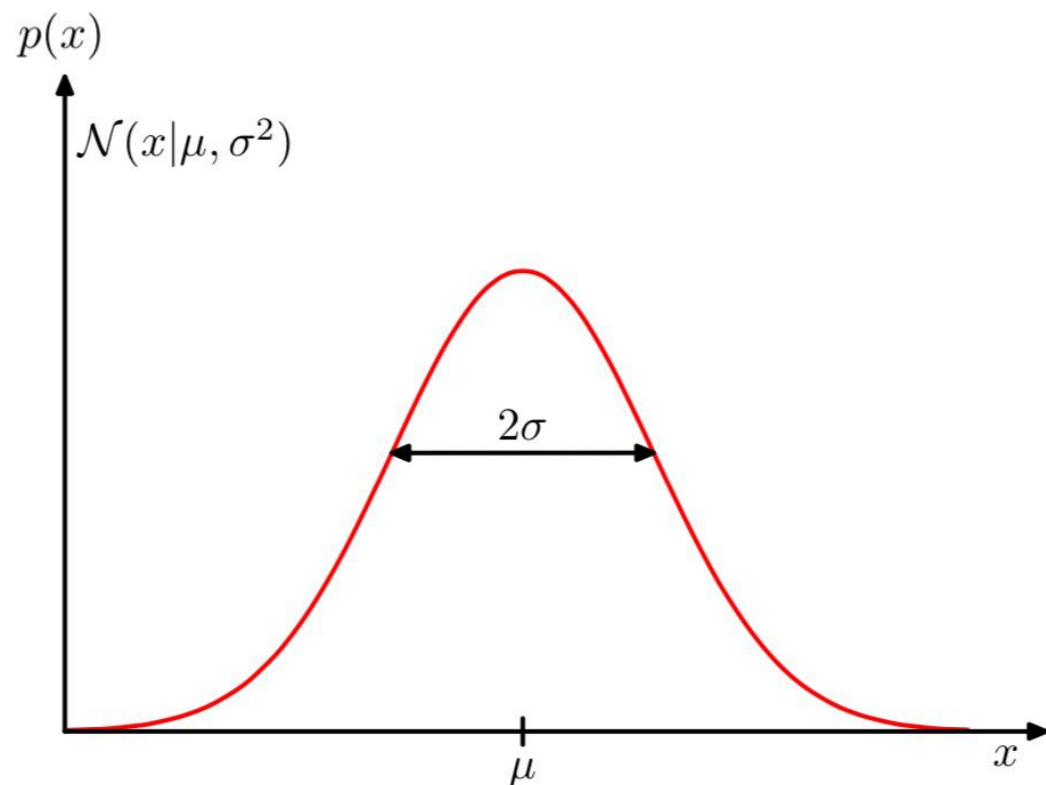
$\beta = \frac{1}{\sigma^2}$: precision

Also called Normal distribution when mean=0 and variance=1

The probability of x to be this value

Probability Theory

- The Gaussian Distribution



$$\mathbb{E}[x] = \int_{-\infty}^{\infty} \mathcal{N}(x|\mu, \sigma^2) x \, dx = \mu$$

$$\mathbb{E}[x^2] = \int_{-\infty}^{\infty} \mathcal{N}(x|\mu, \sigma^2) x^2 \, dx = \mu^2 + \sigma^2$$

$$\text{var}[x] = \mathbb{E}[x^2] - \mathbb{E}[x]^2 = \sigma^2$$

Probability Theory

- The Gaussian Distribution

D-dimensional vector $\mathbf{x}$ of continuous variables

$$\mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{D/2}} \frac{1}{|\boldsymbol{\Sigma}|^{1/2}} \exp \left\{ -\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu}) \right\}$$

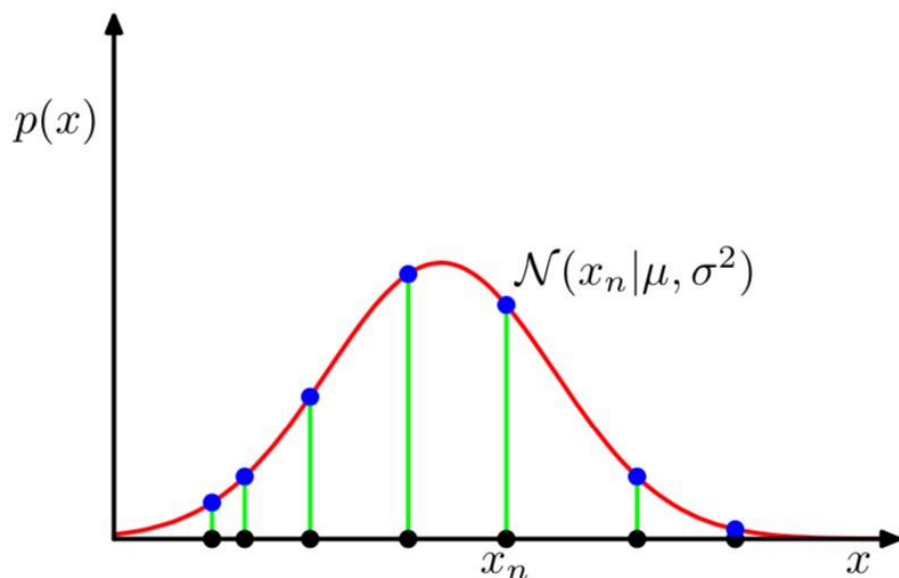
$\boldsymbol{\mu}$ is called the mean

$\boldsymbol{\Sigma}$ is called the covariance

$|\boldsymbol{\Sigma}|$ denotes the determinant of $\boldsymbol{\Sigma}$

Probability Theory

- The Gaussian Distribution



Now suppose that we have a dataset of observations $\mathbf{x} = (x_1, \dots, x_N)^T$, representing N observations of the scalar variable x .

How to determine the mean and variance according to the dataset??

All data points are sampled independently from the same distribution, they are **independent and identically distributed (i.i.d)**.

For i.i.d variables, we can have the **joint probability** as follow

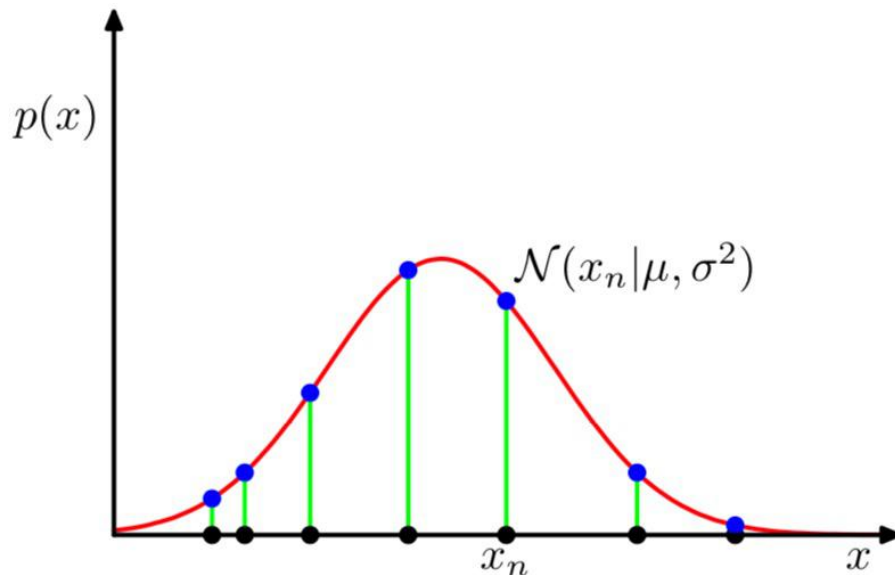
$$p(\mathbf{x}) = p(x_1) p(x_2) p(x_3) \dots p(x_N)$$

So...

$$p(\mathbf{x} | \mu, \sigma^2) = \prod_{n=1}^N \mathcal{N}(x_n | \mu, \sigma^2)$$

Probability Theory

- The Gaussian Distribution



Maximum Likelihood Estimation (MLE)

To find the “best” mean and variance, we can maximise the probability of the parameters given the data

$$p(\mathbf{x} | \mu, \sigma^2) = \prod_{n=1}^N \mathcal{N}(x_n | \mu, \sigma^2)$$

In practice, it is more convenient to maximise the log of the likelihood function. The logarithm is a monotonically increasing function, maximisation of the log of a function is equivalent to maximisation of the function itself

$$\ln p(\mathbf{x} | \mu, \sigma^2) = -\frac{1}{2\sigma^2} \sum_{n=1}^N (x_n - \mu)^2 - \frac{N}{2} \ln \sigma^2 - \frac{N}{2} \ln(2\pi)$$

So.. the **maximum likelihood solution**:

$$\mu_{\text{ML}} = \frac{1}{N} \sum_{n=1}^N x_n \quad \sigma_{\text{ML}}^2 = \frac{1}{N} \sum_{n=1}^N (x_n - \mu_{\text{ML}})^2$$

- Example: Regression - Polynomial Curve Fitting
- Probability Theory
- **Decision Theory**
- Information Theory

Decision Theory

- An example
- Basic intuition
- Minimising the misclassification rate
- Minimising the expected loss
- The reject option
- Loss functions for regression

Decision Theory

- An example

A medical diagnosis problem in which we have taken an X-ray image of a patient, and we wish to determine whether the patient has cancer or not.

We are interested in **the probabilities of the two classes** given the **image**, which are given by $p(C_k|x)$. Using Bayes' theorem, these probabilities can be expressed in the form :

$$p(C_k|x) = \frac{p(x|C_k)p(C_k)}{p(x)}.$$

Note that any of the quantities appearing in Bayes' theorem can be obtained from the joint distribution $p(x, C_k)$ by either marginalizing or conditioning with respect to the appropriate variables.

We can now interpret **$p(C_k)$** as the prior probability for the class C_k , and $p(C_k|x)$ as the corresponding posterior probability. Thus $p(C_1)$ represents the probability that a person has cancer, before we take the X-ray measurement.

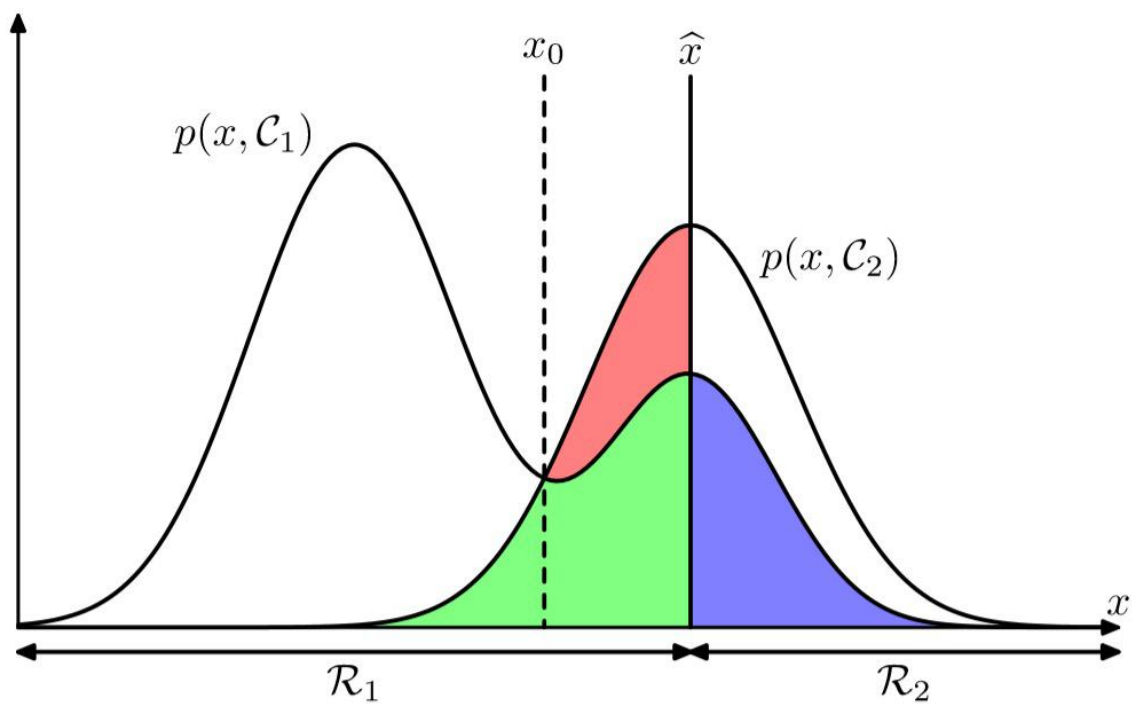
Similarly, $p(C_1|x)$ is the corresponding probability, revised using Bayes' theorem in light of the information contained in the X-ray.

Decision Theory

- Basic intuition

- If our aim is to minimise the chance of assigning x to the wrong class, then intuitively we would choose the class having the higher posterior probability.
- We now show that this intuition is correct, and we also discuss more general criteria for making decisions.

Decision Theory



Decision Theory

- Minimising the expected loss

Loss function, also called a cost function, which is a single, overall measure of loss incurred in taking any of the available decisions or actions.

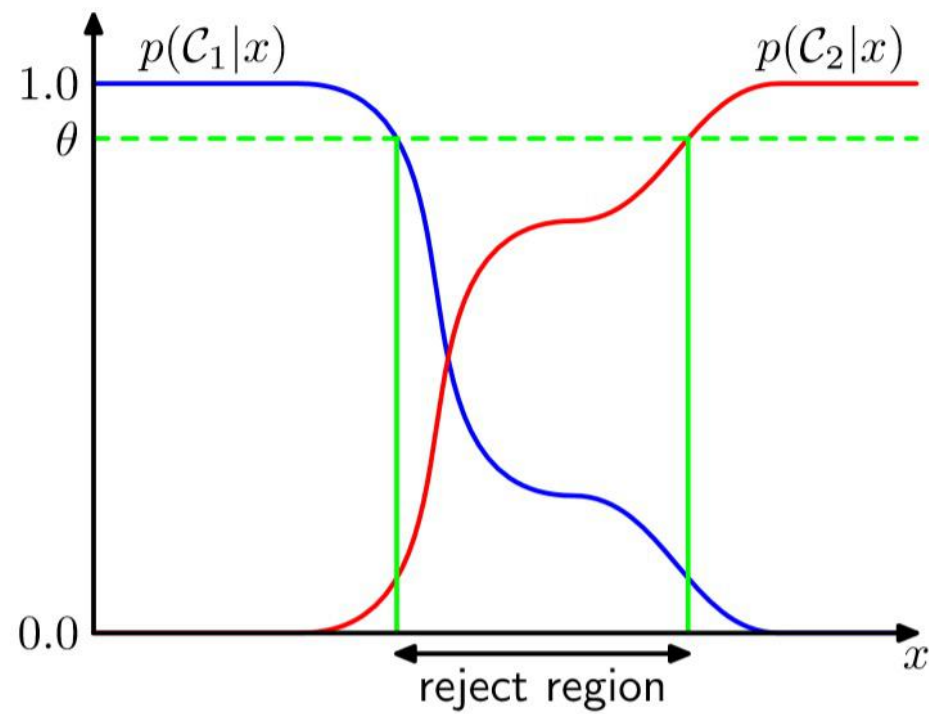
Our goal is then to minimise the total loss incurred.

We seek to minimise the average loss, where the average is computed with respect to this distribution, which is given by

$$\mathbb{E}[L] = \sum_k \sum_j \int_{\mathcal{R}_j} L_{kj} p(\mathbf{x}, \mathcal{C}_k) d\mathbf{x}$$

Decision Theory

- The reject option



- Example: Regression - Polynomial Curve Fitting
- Probability Theory
- Decision Theory
- **Information Theory**

Basic intuition

- Learning that an unlikely event has occurred is more informative than learning that a likely event has occurred.
 - E.g.
 - “the sun rose this morning” : uninformative
 - “there was a solar eclipse this morning”: informative.

Basic intuition

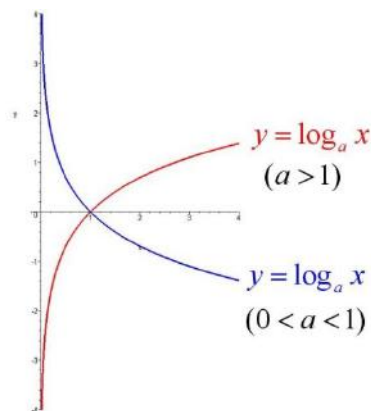
- We would like to quantify information in a way that formalises this intuition. Specifically:
 - Likely events should have low information content, and in the extreme case, events that are guaranteed to happen should have no information content whatsoever.
 - Less likely events should have higher information content.
 - Independent events should have additive information.
 - E.g. Finding out that a tossed coin has come up as heads twice should convey twice as much information as finding out that a tossed coin has come up as heads once.

Self-information

- In order to satisfy all three of these properties, we define the self information of an event x to be

$$I(x) = -\log P(x).$$

(use $\log$ to mean the natural logarithm, with base e . $I(x)$ is therefore written in units of nats)



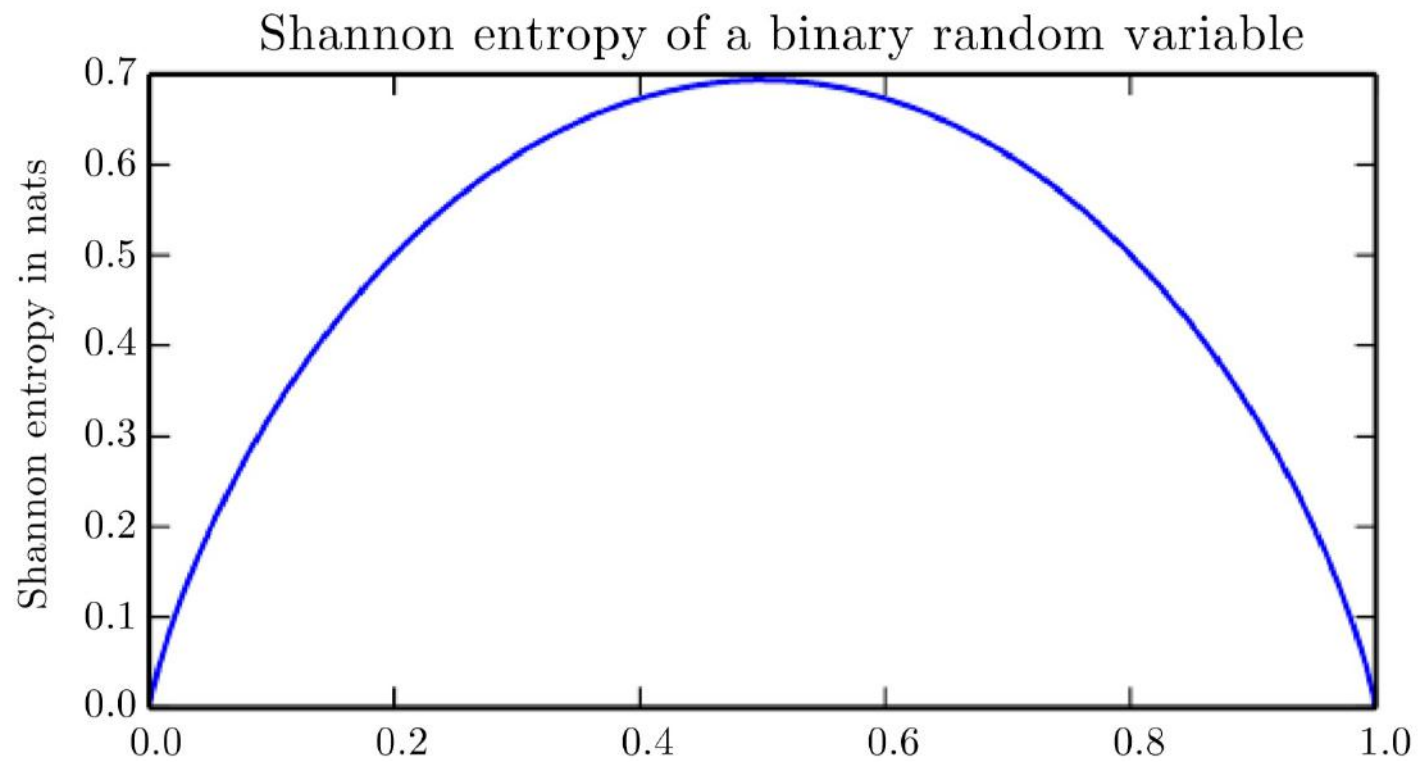
Shannon entropy

- Self-information deals only with a single outcome. We can quantify the amount of uncertainty in an entire probability distribution using the Shannon entropy:

$$H(\mathbf{x}) = \mathbb{E}_{\mathbf{x} \sim P}[I(x)] = -\mathbb{E}_{\mathbf{x} \sim P}[\log P(x)].$$

- The Shannon entropy of a distribution is the expected amount of information in an event drawn from that distribution. It gives a lower bound on the number of bits (if the logarithm is base 2, otherwise the units are different) needed on average to encode symbols drawn from a distribution P .

Example for Shannon entropy



KL divergence

- If we have two separate probability distributions $P(x)$ and $Q(x)$ over the same random variable x , we can measure how different these two distributions are using the Kullback-Leibler (KL) divergence:

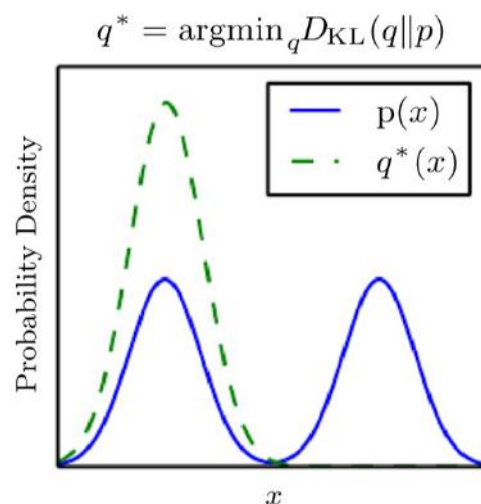
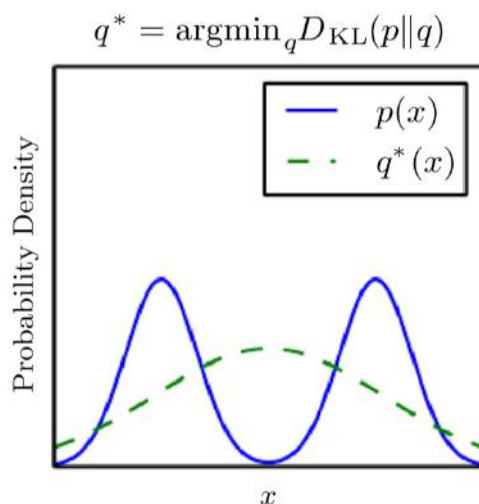
$$D_{\text{KL}}(P\|Q) = \mathbb{E}_{x \sim P} \left[\log \frac{P(x)}{Q(x)} \right] = \mathbb{E}_{x \sim P} [\log P(x) - \log Q(x)] .$$

- In the case of discrete variables, it is the extra amount of information needed to send a message containing symbols drawn from probability distribution P , when we use a code that was designed to minimize the length of messages drawn from probability distribution.

KL divergence

- Because the KL divergence is non-negative and measures the difference between two distributions, it is often conceptualised as measuring some sort of distance between these distributions.
 - Not a true distance measure because it is not symmetric.
 - It is nonnegative.

KL divergence



- Which direction of the KL divergence to use?
 - Some applications require an approximation that usually places high probability anywhere that the true distribution places high probability: left one
 - Other applications require an approximation that rarely places high probability anywhere that the true distribution places low probability: right one

$$D_{\text{KL}}(P||Q) = \mathbb{E}_{\mathbf{x} \sim P} \left[\log \frac{P(\mathbf{x})}{Q(\mathbf{x})} \right] = \mathbb{E}_{\mathbf{x} \sim P} [\log P(\mathbf{x}) - \log Q(\mathbf{x})] .$$

Cross-entropy

- A quantity that is closely related to the KL divergence is the cross-entropy $H(P, Q) = H(P) + D_{KL}(P || Q)$, which is similar to the KL divergence but lacking the term on the left:

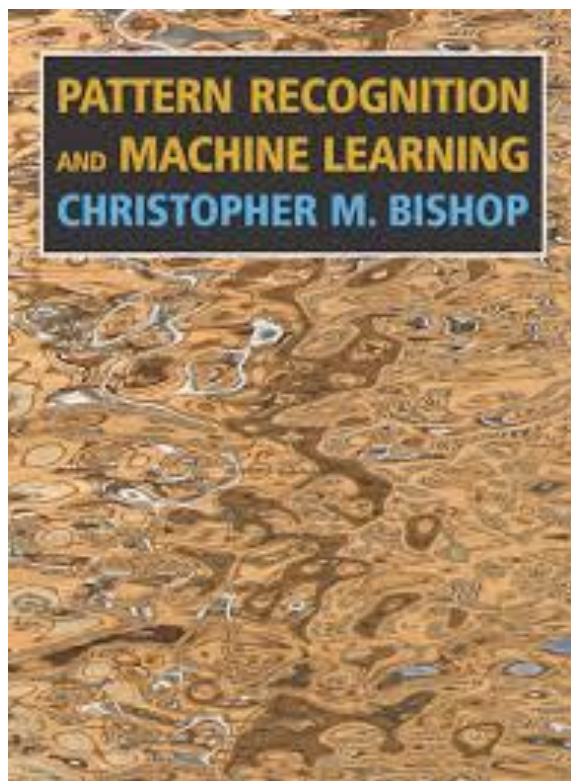
$$H(P, Q) = -\mathbb{E}_{x \sim P} \log Q(x).$$

- Minimising the cross-entropy with respect to Q is equivalent to minimising the KL divergence, because $H(P)$ does not participate in the omitted term.

Mathematic Foundation and Basic Concept

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- Information Theory

Reference



Free Download

<https://github.com/zsdonghao/deep-learning-book/blob/master/All-in-one-pdf>
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Thanks